Polimera Sai Vatsal · B.Tech ECE Student · **saivatsalpolimera@gmail.com** · +91 8309389618

Polimera Sai Vatsal | Resume



- Status: B.Tech ECE, KL University (2025–2029)
- Skills: Strong: C, C++, C#, Python; Basic: JavaScript, .NET; React, Next.js, Node.js, Firebase, Supabase, STM32, ESP32, Git, Docker, FastAPI, Ollama
- Interests: Systems engineering, network security, offline AI, competitive programming
- Activities: Independent engineering projects and competitive programming

Summary

B.Tech ECE student at KL University focused on systems engineering and algorithmic problem-solving in C++. Codeforces Expert (peak 1,676) with 1,500+ problems solved; built projects in network anomaly detection, offline AI, remote access, and full-stack payments. Seeking a software engineering internship.

Projects

2026 **CampusShield AI – Network Threat Detection (SIH 2026)** Python / ML

- Built an Isolation Forest detector for suspicious unidirectional IP flows, explaining which traffic features deviate from normal patterns.
- Built a test ERP and traffic simulator for controlled load generation; integrated Docker, live WebSocket updates with fallback, and replay mode.

Personal **April – Fully Offline Voice Assistant** Python / Ollama

- Built an on-device assistant using Vosk speech-to-text, a local Ollama LLM, FastAPI, and an animated Rive interface.
- Used CPU offload to run a 35B local model under limited VRAM; evaluating a 9B model as a faster alternative.

Personal **E-commerce Store** Next.js / Supabase / Razorpay

- Built a Next.js and Supabase e-commerce store for a family business with Razorpay payment integration.
- Applied Supabase row-level security, IDOR checks, and idempotent payment webhooks.

Personal **ControlDesk – Remote Desktop Application** C# / .NET

- Built a C#/.NET remote desktop app that displays a host screen on a connected device and relays mouse and keyboard input to the host. Repository: ControlDesk.

Personal **FreeGaak – Multi-language Music App** React Native / Expo

- Created a React Native/Expo music app for multi-language discovery, search, playlists, and audio playback. Repository: FreeGaak.

Competitive Programming

- Codeforces Expert (peak 1,676); CodeChef peak 1,744 (Division 2); 1,500+ problems solved. Profiles: Codeforces, CodeChef, and LeetCode.

Education

2025 – 2029 **B.Tech, Electronics and Communication Engineering** KL University